Full-lattice weighted-resolvent dyadic-forest uniform neighbourhood gap

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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Abstract

R-162 proved a cutoff-uniform nonzero neighbourhood only for one recursive pure-dyadic chain. The apparent obstruction to taking all unrelated chains at once was the divergent raw-current sum $\sum_p d_p^2$. R-163 proves that this obstruction is an artefact of taking a norm before the A7 primitive cancellation and before exposing the mandatory recursive edge. On every finite injective dyadic forest, the exact covariance perturbation $T_A T_A^* - \mathrm{Id}$ is expanded into two-sided dyadic words. Every nonidentity word contains at least one coefficient or inserted direction, while its field-current endpoint weight is square summable. The worst current-current weight is $|p|^{-4}$ in dimension three. This gives cutoff-, depth-, and forest-cardinality-uniform covariance-data derivatives through order three. The complete global first-variation connection at the origin vanishes termwise on the nonzero $p\!:\!4p$ Fourier support, so the R-160 4/25 arbitrary-family origin gap transfers. A compact covariance-data box and the complete source, current, trace, projected-force, and sixth-power connections then give one positive analytic radius on which the full expected controller-pullback Hessian is larger than 13/100 in coefficient norm and larger than 1/10 in the recursive tangent metric. A separate exact coefficient audit sharpens the general T-050 reduced-action target: with an absolute anchor and origin-force bound, uniform semiconvexity $\mu > -1/110$ is sufficient; positivity is not necessary. Neither input is proved for random or nonlinear production controls here, so T-050 remains open.

1 Purpose and scope: result and exact boundary

Work on the fixed side-16 three-torus with the hash-pinned A1 symbol, the fixed positive A7 floor, exact nonaliased continuum integration, centered both-sign six-real Gaussian blocks, and one common real-even covariance-matched scalar contraction multiplier. Let $\mathcal{V}$ be any finite set of retained nonzero antipodal momentum classes. Its admitted edge set is

$$\mathcal{E} = \{p : p, 2p \in \mathcal{V}\}. \tag{1.1}$$

There is one raw innovation ξ_p for each vertex. A target that later serves as a source is not duplicated. For deterministic real block matrices A_p define

$$(N_A x)_{2p} = A_p x_p, \qquad r^2 = \|A\|_{\mathfrak{E}}^2 := \sum_{p\in\mathcal{E}} \|A_p\|_{\mathrm{HS}}^2. \tag{1.2}$$

The map $p \mapsto 2p$ is injective, so (1.1) is a disjoint union of finite nonbranching chains. It is called the pure-dyadic forest below; it is not a general visit forest.

Put

$$T_A = (\mathrm{Id} - N_A)^{-1}, \qquad B_A = T_A - \mathrm{Id}, \qquad \eta = T_A\xi, \qquad h = B_A\xi. \tag{1.3}$$

Thus every later controller reads the actual shifted state of its parent.

R-163. There is one explicit analytic number $\delta_{\mathrm{F}} > 0$, independent of the finite cutoff, the number of unrelated chains, every chain depth and starting mode, and the admitted common multiplier, such that $\|A\|_{\mathfrak{E}} < \delta_{\mathrm{F}}$ implies, for every $H \neq 0$,

$$\boxed{D^2(\mathcal{A} \circ \chi)_A[H, H] > \frac{13}{100}\|H\|_{\mathfrak{E}}^2 > \frac{1}{10}\mathcal{G}_A(H, H).} \tag{1.4}$$

Here $\chi(A) = B_A\xi$,

$$\mathcal{G}_A(H, H) = \mathbb{E}\|D\chi_A[H]\|^2, \tag{1.5}$$

and $\mathcal{A}$ is the one globally summed R-141 production scalar with source cost, complete A7 endpoint/current owner, and terminal sixth power included once. Equation (1.4) is an expected global controller-pullback theorem, not an intrinsic physical-source or pathwise conditional theorem.

The result does not cover random or nonlinear past-dependent coefficients, parallel edges, branching visits, revisits or cycles, a zero mode, aliasing, nonzero mean, mismatched regulators, floor or cutoff removal, T-050, A13, Nelson, an interacting measure, any phase or PDE, or Sector-A closure.

2 Forest resolvent, source cost, and tangent metric

The finite shift N_A is nilpotent. For $r < 1$ and $\tau = (1 - r)^{-1}$,

$$\|N_A\|_{\mathrm{op}} \leq \|N_A\|_{\mathrm{HS}} = r, \quad \|T_A\|_{\mathrm{op}} \leq \tau, \quad \|B_A\|_{\mathrm{HS}} \leq r\tau. \tag{2.1}$$

Exact resolvent differentiation gives

$$D^m T_A[H_1, \ldots, H_m] = \sum_{\sigma \in S_m} T_A N_{H_{\sigma(1)}} T_A \cdots N_{H_{\sigma(m)}} T_A, \tag{2.2}$$

and hence

$$\|D^m T_A[H_1, \ldots, H_m]\|_{\mathrm{HS}} \leq m!\tau^{m+1} \prod_{j=1}^{m} \|H_j\|_{\mathfrak{E}}, \qquad m = 1, 2, 3. \tag{2.3}$$

No matrix dimension or path count enters (2.3).

The exact source cost is

$$\mathcal{A}_{\mathrm{CM}}(A) = \frac{9}{20}\|B_A\|_{\mathrm{HS}}^2. \tag{2.4}$$

For $r \leq 1/2$, the complete third derivative, including the source projected-force connection, obeys the R-162 bound

$$\|D^3\mathcal{A}_{\mathrm{CM}}(A)\|_{\mathrm{op}} \leq \frac{27}{5}(1 + r)\tau^5 \leq \frac{1296}{5}. \tag{2.5}$$

Moreover

$$(1 + r)^{-4}\|H\|_{\mathfrak{E}}^2 \leq \mathcal{G}_A(H, H) \leq (1 - r)^{-4}\|H\|_{\mathfrak{E}}^2. \tag{2.6}$$

These are forest estimates because the direct-sum shift still satisfies the same Hilbert-Schmidt and operator bounds.

3 Origin transfer and the complete connection

At $A = 0$,

$$D\chi_0[H] = N_H\xi, \qquad D^2\chi_0[H, K] = (N_H N_K + N_K N_H)\xi. \tag{3.1}$$

The $4p$ component of the acceleration is exactly

$$(H_{2p}K_p + K_{2p}H_p)\xi_p. \tag{3.2}$$

There are no cross-chain acceleration terms because doubling has a unique parent and child. Each term in (3.2), paired with its original carrier, has only the nonzero exact-torus Fourier support

$$\{\pm 3p, \pm 5p\}. \tag{3.3}$$

The complete uncontrolled covariance data are stationary, so the first derivative of every globally integrated A7 and sixth-power covariance scalar at the origin is spatially constant. Therefore its pairing with (3.3) vanishes term by term. The source connection also vanishes because $B_0 = 0$. Another raw mode at $3p$ or $5p$ does not alter this linear connection: innovations are independent and the origin data remain stationary. Such factor-three relations do occur in the quadratic tangent term, and R-160 already includes them in the complete affine root-union Hessian.

Equivalently, with $P_A = T_A T_A^*$,

$$D^2 P_0[H, H] = 2N_H N_H^* + 2N_H^2 + 2(N_H^*)^2. \tag{3.4}$$

The last two terms are precisely the cancelled connection directions. The first term is the R-160 one-shot family with target-source reuse. Hence

$$\boxed{D^2(\mathcal{A} \circ \chi)_0[H, H] > \frac{4}{25}\|H\|_{\mathfrak{E}}^2.} \tag{3.5}$$

This is a full-action statement. No local covariance-normal atom is identified with P_{comp}.

4 The two-sided weighted dyadic-word lemma

The raw stacked-current synthesis is not cutoff-uniformly bounded. The repair is to retain $P_A - \mathrm{Id}$ before taking a norm:

$$P_A = \sum_{a,b \geq 0} N_A^a (N_A^*)^b. \tag{4.1}$$

Let E, F be value or stacked-current syntheses, and let e_p, f_p be their block operator-norm envelopes. Suppose

$$\left(\sum_p e_{2^a p}^2 f_{2^b p}^2\right)^{1/2} \leq \Gamma_{EF} q_E^a q_F^b. \tag{4.2}$$

For

$$Y_{EF}(A) = E(P_A - \mathrm{Id})F^*, \tag{4.3}$$

the exact path expansion gives, for $m = 0, 1, 2, 3$,

$$\boxed{\|D^m Y_{EF}(A)[H_1, \ldots, H_m]\| \leq \Gamma_{EF} \frac{d^m}{dr^m}\left\{\frac{1}{(1 - q_E r)(1 - q_F r)} - 1\right\} \prod_{j=1}^m \|H_j\|_{\mathfrak{E}}.} \tag{4.4}$$

For $m = 0$ the empty product is one. For $m \geq 1$ the derivative kills the subtracted identity.

To prove (4.4), fix one word (a, b) of degree $n = a + b$. Its labelled mth derivative has at most

$$(n)_m = n(n-1)\cdots(n-m+1) \tag{4.5}$$

terms. Matrix submultiplicativity reduces each term to one coefficient $e_{2^a p} f_{2^b p}$, m inserted direction sequences, and $n - m$ coefficient sequences. Holder with exponents $2, 2n, \ldots, 2n$, together with $\ell^2 \subset \ell^{2n} \subset \ell^\infty$ and injectivity of every $p \mapsto 2^j p$, gives

$$\sum_p e_{2^a p} f_{2^b p}, |\text{path monomial}| \leq \Gamma_{EF} q_E^a q_F^b r^{n-m} \prod_j \|H_j\|_{\mathfrak{E}}. \tag{4.6}$$

The same proof permits repeated appearances of an edge on the two sides of the word. Summing (4.6) over (a, b) gives (4.4). Fourier collisions are paid by the absolute geometric sum; no false cross-chain orthogonality is used.

4.1 Exact A1 full-lattice weights

Write $x_p = |p|^2$. R-161 supplies

$$s_p^2 \le \frac{g}{1+x_p^2}, \qquad d_p^2 \le \frac{g x_p}{1+x_p^2}, \qquad x_p > \frac{3}{20}|n|^2 =: c_0|n|^2, \tag{4.7}$$

where

$$g = \frac{244140625000000000}{288000000000947494031}. \tag{4.8}$$

The three-dimensional lattice bounds are

$$\sum_{n\neq 0} |n|^{-4} < 52, \qquad \sum_{n\neq 0} |n|^{-6} < 35, \qquad \sum_{n\neq 0} |n|^{-8} < 32. \tag{4.9}$$

Consequently (4.2) holds with

$$\begin{array}{c|c|c} EF & \Gamma_{EF} & (q_E, q_F) \\ \hline SS & \sqrt{32}\, g c_0^{-2} & (1/4, 1/4) \\ DD & \sqrt{52}\, g c_0^{-1} & (1/2, 1/2) \\ SD & \sqrt{35}\, g c_0^{-3/2} & (1/4, 1/2). \end{array} \tag{4.10}$$

The worst DD square weight is $|p|^{-4}$, which is summable in dimension three. By contrast, the forbidden raw bound uses $\sum_p d_p^2 \sim \sum_p |p|^{-2}$ and diverges. Thus the recursive edge and the subtraction $P_A - \mathrm{Id}$ are both load-bearing.

5 Covariance data and the one global scalar

The exact covariance perturbation is

$$R_A = P_A - \mathrm{Id} = B_A + B_A^* + B_A B_A^*. \tag{5.1}$$

Primitive A7 cancellation is performed algebraically before norms:

$$\begin{aligned} C_A &= C_0 + S R_A S^*, \\ U_A := Q_A - Q_0 &= \sum_{i=1}^{3} D_i R_A D_i^*, \\ K_{A,i} &= S R_A D_i^*. \end{aligned} \tag{5.2}$$

Common evenness gives $K_0 = 0$. Applying (4.4), followed only by the fixed three-current contraction and fixed six-real output-rank inequalities, gives finite constants J_m, $m = 0, 1, 2, 3$, such that on $r \le 1/2$,

$$\|D^m Y_A[H_1, \dots, H_m]\|_{\mathcal{Z}} \le J_m \prod_j \|H_j\|_{\mathfrak{E}}, \tag{5.3}$$

where

$$Y_A = (C_A, U_A, K_A), \qquad \|(C, U, K)\|_{\mathcal{Z}}^2 = \|C\|_{\mathrm{op}}^2 + \|U\|_{\mathrm{op}}^2 + \sum_i \|K_i\|_{\mathrm{HS}}^2. \tag{5.4}$$

The primary and independent executables construct conservative exact J_m from (4.10); no decimal approximation is used by the theorem.

The globally summed endpoint/current owner and terminal sixth moment combine into the R-162 pointwise covariance scalar

$$\begin{aligned} \Phi(C, U, K) = \frac{1}{2}\Big\{ &\mathrm{Tr}[U \overline{\mathcal{B}}(C)] + \sum_i K_i \otimes K_i : \overline{\mathcal{T}}(C) \Big\} \\ &+ \frac{3}{20}\Big[(\mathrm{Tr}\, C)^3 + 6\,\mathrm{Tr}\, C\ \mathrm{Tr}(C^2) + 8\,\mathrm{Tr}(C^3)\Big]. \end{aligned} \tag{5.5}$$

Up to the fixed zero-control constant, the complete expected action is

$$F(A) = \frac{9}{20}\|B_A\|_{\mathrm{HS}}^2 + \int_{\mathbb{T}_{16}^3} \Phi(Y_A(x))\,dx. \tag{5.6}$$

Equation (5.6) contains every forward block and its legal reverse as one self-adjoint Hessian, their balanced symmetric cross once, and every projected-force connection once. Derived states are not independent low variables. The sixth power remains inside Φ: after nonlinear pullback its connection is not positive by itself.

The R-161 bound on C_0 and (5.3) put every $Y_A(x)$ in one compact finite-dimensional covariance-data box. At the positive A7 floor, Gaussian heat differentiation gives finite constants

$$M_j = \sup_{Y \text{ in the box}} \|D^j\Phi(Y)\| < \infty, \qquad j = 1, 2, 3. \tag{5.7}$$

No inverse covariance is used, so singular admitted multipliers are allowed.

6 Uniform modulus and retained forest gap

The complete third-order chain rule is

$$\|D^3(\Phi \circ Y_A)\| \le M_3 J_1^3 + 3M_2 J_1 J_2 + M_1 J_3. \tag{6.1}$$

The last term is mandatory because the recursive covariance has a nonzero third jet. With torus volume $16^3 = 4096$, define

$$L_{\mathrm{F}} := \frac{1296}{5} + 4096\left(M_3 J_1^3 + 3M_2 J_1 J_2 + M_1 J_3\right). \tag{6.2}$$

Equations (2.5), (5.3), and (6.1) give

$$\|D^2F(A) - D^2F(0)\|_{\mathrm{op}} \le L_{\mathrm{F}}\|A\|_{\mathfrak{E}}, \qquad \|A\|_{\mathfrak{E}} \le \frac{1}{2}. \tag{6.3}$$

Choose

$$\boxed{\delta_{\mathrm{F}} := \min\left\{\frac{1}{2}, \frac{3}{100(1+L_{\mathrm{F}})}\right\} > 0.} \tag{6.4}$$

Then (3.5) and (6.3) imply

$$D^2F(A)[H,H] > \left(\frac{4}{25} - \frac{3}{100}\right)\|H\|_{\mathfrak{E}}^2 = \frac{13}{100}\|H\|_{\mathfrak{E}}^2. \tag{6.5}$$

Since $\delta_{\mathrm{F}} < 3/100$ and

$$\left(\frac{100}{97}\right)^4 < \frac{13}{10}, \tag{6.6}$$

the upper bound in (2.6) turns (6.5) into

$$D^2F(A)[H,H] > \frac{1}{10}\mathcal{G}_A(H,H), \tag{6.7}$$

which proves R-163.

7 Exact T-050 semiconvexity threshold

This section sharpens the target for the still-missing general production estimate. It does not assert that its hypotheses hold beyond R-163.

Let $X(h) = \mathbb{E}\sum_j \|h_j\|_2^2$ and let $Y_6(h)$ denote the terminal sixth power. Suppose a complete reduced action $\mathcal{A}_{\mathrm{red}}$, after any genuine low coordinate has been eliminated by its exact range-compatible Schur complement, obeys along every radial segment

$$D^2\mathcal{A}_{\mathrm{red}}(th)[h,h] \geq \mu X(h), \qquad 0 \leq t \leq 1, \tag{7.1}$$

and suppose

$$\mathcal{A}_{\mathrm{red}}(0) \geq -C_0, \qquad \|D\mathcal{A}_{\mathrm{red}}(0)\|_*^2 \leq B_g. \tag{7.2}$$

Taylor's formula and Young's inequality give, for every $\eta > 0$,

$$\mathcal{A}_{\mathrm{red}}(h) \geq -C_0 - \frac{B_g}{4\eta} + \left(\frac{\mu}{2} - \eta\right) X(h). \tag{7.3}$$

Because the explicit action is

$$\mathcal{A}_{\mathrm{red}}(h) = \mathbb{E}V_J^{\mathrm{ren}}(Z_h) + \frac{9}{20}X(h) + \frac{3}{20}Y_6(h), \tag{7.4}$$

equation (7.3) yields

$$\mathbb{E}V_J^{\mathrm{ren}}(Z_h) \geq -\epsilon_v X(h) - \frac{3}{20}Y_6(h) - C, \tag{7.5}$$

with

$$\epsilon_v = \frac{9}{20} - \frac{\mu}{2} + \eta. \tag{7.6}$$

For the registered $p = 11/10$ T-050 gate,

$$\frac{1}{2p} = \frac{5}{11}, \qquad \frac{5}{11} - \frac{9}{20} = \frac{1}{220}. \tag{7.7}$$

Therefore

$$\boxed{\mu > -\frac{1}{110}} \tag{7.8}$$

is sufficient: choose $0 < \eta < 1/220 + \mu/2$. Then

$$\epsilon_v < \frac{5}{11}, \qquad \epsilon_6 = \frac{3}{20} < \frac{27}{100}. \tag{7.9}$$

If the explicit $9/10$ source Hessian is separated, (7.8) asks the complete adverse owner only for

$$K_{\mathrm{owner}}^{\mathrm{red}} \succ -\frac{10}{11}G_{\mathrm{src}}, \tag{7.10}$$

with a strict margin, not the stronger previously emphasized $-4G_{\mathrm{src}}/5$ bound. Equations (7.2) remain indispensable. Thus R-163 narrows the quantitative target but does not close the absolute anchor, origin force, random law, or pathwise conditional gates.

If a genuine independent low variable is present, its block must satisfy the Douglas range condition before the Moore–Penrose Schur complement is used. If low is a nonlinear graph, (7.1) must instead use the full pullback Hessian

$$D\chi^* H_{\mathrm{prod}} D\chi + \langle g_{\mathrm{prod}}, D^2\chi\rangle. \tag{7.11}$$

Forward, legal reverse, balanced, and low are never separate reserves.

8 Failed shortcuts and reusable boundaries

1. **Positive local gaps do not aggregate.** The complete one-use matrix

$$\begin{pmatrix} 3/20 & -1/5 \\ -1/5 & 3/20 \end{pmatrix} \tag{8.1}$$

has both diagonal gaps above $1/10$ but eigenvalues $-1/20$ and $7/20$. The cross is already forward plus its legal reverse, counted once.

2. **A low diagonal does not repair a kernel cross.** The matrix

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \tag{8.2}$$

has determinant -1. Range compatibility is necessary.

3. **Raw-current boundedness is false.** The estimate $\|DB\|_{\mathrm{HS}} \le \|D\| \|B\|_{\mathrm{HS}}$ is not cutoff uniform. The proof of (4.4) differentiates the block words directly; inserted directions provide the required ℓ^2 factors even when differentiation removes every literal A factor.

4. **Cross-chain Fourier orthogonality is false.** Factor-three and other collisions occur. Equations (4.2)–(4.6) use an absolute geometric sum and therefore do not need orthogonality.

5. **Convexity of the sixth power is insufficient after pullback.** The connection $D\Phi[D^3Y]$ and the chart-acceleration term in (7.11) remain inside the one global scalar.

6. **Random nonlinear coefficients leave the Gaussian chart.** For example, a coefficient proportional to $\operatorname{sgn}\xi$ produces $|\xi|$. Covariance data no longer determine the endpoint law, so (5.5) cannot be reused.

9 Devil's-advocate audit

- **Objection: a hidden forest-cardinality factor remains. DISMISSED.** Each two-sided word uses Holder against the full-lattice square weight (4.2), and the word lengths are summed geometrically.

- **Objection: common-ancestor BB^* terms were omitted. DISMISSED.** They are exactly the words $a, b \ge 1$ in (4.1).

- **Objection: another chain at $3p$ spoils the origin connection. DISMISSED under the declared stationary hypotheses.** The connection is linear in the off-diagonal $p : 4p$ covariance second jet and pairs with a spatially constant origin derivative. The quadratic factor-three cross is already in R-160.

- **Objection: the divergent primitive current covariance was hidden. DISMISSED.** Equation (5.2) forms $Q_A - Q_0$ before every norm. Raw Q_A is never bounded.

- **Objection: the theorem covers a general predictable forest. UPHELD AS PROHIBITED.** The graph is an injective disjoint union of dyadic chains with deterministic matrices. General visits and random laws remain open.

- **Objection: (7.8) closes T-050. UPHELD AS PROHIBITED.** It is a sufficient threshold. The production conditional semiconvexity, anchor, and origin-force bounds are not supplied.

- **Objection: the result selects BCC or another phase/PDE. UPHELD AS PROHIBITED.** R-163 is phase-neutral source-action mathematics and contains no candidate-state comparison.

External review is invited for the one-vertex-per-mode convention, antipodal and six-real normalization, origin stationarity pairing, matrix-valued Holder step, repeated-edge two-sided words, exact A1 exponents, current contraction, primitive cancellation, complete third chain rule, tangent-metric comparison, and the $-1/110$ threshold algebra.

10 Reproduction and Result footer

Dependencies: A1, A7; R-107, R-141, R-155, R-160, R-161, R-162.

Reproduction command: primary executable below.

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_full_lattice_weighted_resolvent_dyadic_forest_uniform_
neighborhood_gap_boundary.py
```

Independent command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_full_lattice_weighted_resolvent_dyadic_forest_uniform_
neighborhood_gap_boundary_independent.py
```

Integrated command:

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
a13_classii_full_lattice_weighted_resolvent_dyadic_forest_uniform_
neighborhood_gap_boundary_verify.py
```

```
Result ID: A13-CLASSII-FULL-LATTICE-WEIGHTED-RESOLVENT-DYADIC-
 FOREST-UNIFORM-NEIGHBORHOOD-GAP-BOUNDARY; Ledger ID: R-163.
Precise statement: one positive analytic l2(HS) radius, uniform over every
 finite injective deterministic pure-dyadic forest, retains the complete
 expected controller-pullback gap >13/100>1/10; separately, reduced-action
 semiconvexity mu>-1/110 plus anchor and force is sufficient for T-050.
Scope: side-16 d=3 A1/A7, fixed positive floor, exact torus, centered
 independent Gaussian blocks, common-even matched multiplier, deterministic
 injective dyadic forest.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Expected output: primary, independent, integrated, PDF and release PASS.
Falsification gate: any forest, word, origin, owner, threshold, hash, PDF,
 exploration, public-surface, or release assertion fails.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no intrinsic/random/nonlinear/branching/revisit/
 pathwise-conditional/removal/T-050/A13/Nelson/measure/phase/PDE/Sector-A
 theorem.
Proof complete: false; T-050 closed: false; Sector A closed: false.
Next required action: test the owner-complete -10/11 threshold, absolute
 anchor, and origin force on a legal production conditional chart.
```